



AMITY UNIVERSITY

MADHYA PRADESH

Established vide Government of Madhya Pradesh Act No. 27 of 2010

Syllabus

Programme Name: B. Tech. (CSE)		Session: 2023-27
Course Code: IT601	Course Name: COMPUTER NETWORKS	Semester: VI

Credits (Total)	L	T	P	Marks (Internal/External)		Contact Hours (per week)	Independent Study Hour (per week)	Section (Group)
3	2	1	0	30	70	3	3	
UG level						Basic and applied	Student specific course outcome	Higher Education Placement Research

Course Objective:

The objective of the course is to provide introductory concepts in Communications such as Signaling, Encoding, Modulation, Error Detection & Correction. The course is also aimed at providing basic understanding of Computer networks starting with OSI Reference Model, Protocols at different layers with special emphasis on IP, TCP & UDP and Routing algorithms. Students are expected to learn applications of Computer Networks and interfacing standards along with current edge of the network security techniques.

Course outcomes: After completion of course, the student will be able to:

CO-1	Show clear understanding of the basic concepts of data communications including the key aspects of networking and their interrelationship, packet switching, circuit switching and cell switching as internal and external operations, physical structures, types, models, and internetworking.
CO-2	Able to intelligently compare and contrast local area networks and wide area networks in terms of characteristics and functionalities. Able to identify limitations of typical communication systems.
CO-3	Able to differentiate among and discuss the four levels of addresses (physical, logical, port, and specific used by the Internet TCP/IP protocols.
CO-4	Able to understand applications and security issues of computer networks

Teaching Pedagogy:

T1	Classroom teaching (white board), Power Point Presentations, Interactive lectures, Inquiry based teaching
T2	Class Test, Assignments, Flip Class/ Seminars, Quiz, Oral Viva-voce examination

Assessment Tools

AT1-1	Quiz/Class Test
AT1-2	Activity Based Learning
AT1-3	Midterm Exams
AT1-4	Flip Class
AT1-5	Seminar Presentation
AT1-6	Assignments
AT1-7	Poster
AT1-8	Oral Viva-voce examination
AT1-9	Industrial Visit Report

Prerequisites: computer science fundamentals, networking basics, mathematics (especially discrete mathematics), digital logic, operating systems, programming skills, and basic electronics knowledge

Module wise contents details	Assessment tools
Module I: Introduction to Data Communications (9 Hours) Introduction to Data Communication, Networks-protocols, advantages, disadvantages & applications, Line Configuration, topology, Transmission mode, Classification of networks. OSI and TCP IP Model. Parallel & Serial Transmissions, Analog & Digital Signals, Periodic & Aperiodic Signals, Modulation---Amplitude Modulation, Frequency Modulation, Phase Modulation, Pulse Amplitude Modulation, Pulse Code Modulation, Sampling.	Quiz/Class Test Mid-term Exam Assignment
Module II: Data Link Layer (9 Hours) Framing, Line Discipline, Types of Errors, Error Detection & Correction (VRC, LRC, CRC, Checksum, Hamming Code), Flow Control (Stop-and-wait & Sliding Window), Error Control (Stop & Wait ARQ, Sliding Window ARQ using Go-back n method and Selective-Reject). CSMA/CD, Project 802, IEEE Standards-802.3, Token Bus (802.4), Token Ring (802.5), Introduction to Bridges.	Quiz/Class Test Mid-term Exam Assignment
Module III: Network and Transport Layer (9 Hours) Network Layer in Internet-IP protocol, IP addressing, Subnetting & Internet Control Protocols, Transport Layer in Internet-TCP & UDP, Internal Organization of Network Layer, Routing Algorithms-Shortest Path Routing, Flooding, Distance Vector Routing, Link State Routing, Transport Layer Design issues, General Principles of Congestion, Congestion Prevention Policies, Duties of Transport Layer, Connection Establishment & Connection Termination.	Quiz/Class Test Mid-term Exam Oral Viva-voce examination Seminar
Module IV: Application Layer (9 Hours) Application Layer Protocols – Domain name space, DNS in internet, web server, electronic mail, FTP, WWW, HTTP, SMTP, POP, IMAP, SNMP, DHCP, NFS, NTP, Telnet, SSH.	Quiz/Class Test Mid-term Exam Oral Viva-voce examination Presentation
Module V: Network Security (9 Hours) Introduction to the Concept of Security, SSL, Cryptographic Techniques in Network Security, Computer-based Symmetric and Asymmetric Key Cryptographic Algorithms, Public Key Infrastructure (PKI), Internet Security Protocols, Network Security design paradigm.	Quiz Assignment Industrial Visit Report Poster Oral Viva-voce examination

Additional Learning:

List of Assignments	• Explain Http and WWW in details. • Differentiate OSI and TCP/IP. • Differentiate IPV4 and IPV6. • What are the types of cryptography explain with diagram? • Explain Hamming Code in data link layer • Differentiate unicast, multicast, anycast in IPV6 • Explain the Techniques to improve QoS. • Explain the flow control techniques in data link layer with suitable diagram. • What is DNS? What is the need of DNS explain with its working diagram? • Explain the ISDN services and ATM Concept. • Differentiate LAN and VLAN • Differentiate IEEE 802.11 and IEEE 802.15a • What do you understand by Cellular telephony and Satellite network? • Explain the concept of SONET.
Suggested reading:	• Behrouz., Forouzan., “Data Communication and Networking”, TMH • W. Stallings, "Data and Computer Communication" PHI • A.S. Tanenbaum, "Computer Networks" , PHI • Kennedy, “Electronics Communication System”, TMH. • Kurose and Rose – “Computer Networking -A top down approach featuring the internet” – Pearson Education • Leon, Garica, Widjaja – “Communication Networks” – TMH • Walrand – “Communication Networks” – TMH.
Suggested e-resources (Websites/e-books)	3. Online Courses and Tutorials: • Coursera: "Computer Networking" by Stanford University. • edX: "Introduction to Computer Networking" by NYU. • Udemy: Various courses on networking fundamentals and advanced topics. • NPTEL: Various courses on networking fundamentals and advanced topics. 4. Books: • "Computer Networking: A Top-Down Approach" by James F. Kurose and Keith W. Ross. • "Data Communications and Networking" by Behrouz A. Forouzan. • "TCP/IP Illustrated" by Kevin R. Fall and W. Richard Stevens. 5. Websites and Documentation: • Cisco Networking Academy: Offers various free resources, including courses and interactive learning tools. • Wireshark: A network protocol analyzer that can be used for learning and analyzing network traffic. • Internet Engineering Task Force (IETF): Provides RFCs (Request for Comments) which are documents that define the standards and protocols used in computer networking. • The TCP/IP Guide: An online resource providing comprehensive information on TCP/IP protocols and networking concepts.

	6. Online Tools: • Packet Tracer: A network simulation tool developed by Cisco for designing, configuring, and troubleshooting network scenarios. • GNS3 (Graphical Network Simulator-3): A network emulator that allows users to simulate complex networks. 7. Online Forums and Communities: • Stack Overflow: A community where you can ask questions related to networking and data communication. • Reddit: Subreddits like r/networking and r/NetworkingForBeginners can be helpful for discussions and learning from others' experiences.
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Assessment Plan:

Component of Evaluation	Description	Code	Weightage %
Continuous Internal Evaluation	Mid Term	CT	15%
	Seminar/Viva-Voce/Quiz/Home Assignment	S/V/Q/HA	10%
Attendance	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The dispensation of 25% includes all types of leaves. including medical leaves.	A	5%
End Semester Examination	End Semester Examination	ESE	70%
Total			100%

Abbreviations: CT: Class Test, HA: Home Assignment, S/V/Q: Seminar/Viva/Quiz, ESE: End Semester Examination; A: Attendance

Course Articulation Matrix (Mapping of COs with POs)

	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO1	3	3	3	2	3			2	2	3		1	2	3	2
CO2	3	3	3	3	1				2	2			3	2	1
CO3	3	3	3	2	3	1		2	3	2	2	1	3	2	2
CO4	3	3	3	3	2			1	2	2		2	3	3	3
CO5	3	3	3	3	2	1		1	2	2	1	2	2	2	2

1: strongly related, 2: moderately related and 3: weakly related